

The `toolAliases` parameter allows you to create custom names for tools, primarily used to resolve naming conflicts between MCP servers or create more intuitive names for complex tool identifiers. It accepts a JSON object where keys are the original tool names (including MCP server prefixes like `@server_name/tool_name`) and values are the new alias names you want to use (without server prefixes).

This parameter is particularly useful when multiple MCP servers provide tools with the same name (e.g., both `@github-mcp/get_issues` and `@gitlab-mcp/get_issues` can be aliased to `github_issues` and `gitlab_issues`), or when you want to create shorter, more memorable names for frequently used tools with long identifiers, while the aliased tools still respect all security controls defined in `allowedTools` and `toolsSettings` using their original names.

Objective

Learn how to use `toolAliases` to create user-friendly names for complex MCP tool names.

Instructions

Step 1: Create Agent with Tool Aliases

```
1  cat > .kiro/agents/aliased-aws-agent.json << 'EOF'
2  {
3    "name": "aliased-aws-agent",
4    "description": "AWS agent with user-friendly tool aliases",
5    "prompt": "You are an AWS assistant with easy-to-use tool names.",
6    "mcpServers": {
7      "aws-knowledge-mcp-server": {
8        "command": "uvx",
9        "args": [
10         "mcp-proxy",
11         "--transport",
12         "streamablehttp",
13         "https://knowledge-mcp.global.api.aws"
14       ],
15       "timeout": 30000
16     },
17     "awslabs.aws-pricing-mcp-server": {
18       "command": "uvx",
19       "args": ["awslabs.aws-pricing-mcp-server@latest"],
20       "env": {
21         "FASTMCP_LOG_LEVEL": "ERROR",
22         "AWS_REGION": "us-east-1"
23       },
24       "timeout": 30000
25     }
26   },
27   "tools": [
28     "read",
29     "@aws-knowledge-mcp-server",
30     "@awslabs.aws-pricing-mcp-server"
31   ],
32   "allowedTools": [
33     "read",
34     "@aws-knowledge-mcp-server/aws___search_documentation",
35     "@awslabs.aws-pricing-mcp-server/get_pricing"
36   ],
37   "toolAliases": {
38     "@aws-knowledge-mcp-server/aws___search_documentation": "aws_search",
39     "@awslabs.aws-pricing-mcp-server/get_pricing": "pricing_lookup"
40   }
41 }
42 EOF
```

Step 2: Test Tool Aliases

1. Start custom agent

```
1  kiro-cli chat --agent aliased-aws-agent
```

2. Check mcp server

```
1  /mcp
```

Expected:

```
[aliased-aws-agent] > /mcp

awslabs.aws-pricing-mcp-server
✓ awslabs.aws-pricing-mcp-server loaded in 9.84 s

aws-knowledge-mcp-server
✓ aws-knowledge-mcp-server loaded in 10.50 s
```

3. Check tools

```
1  /tools
```

Expected:

```
[aliased-aws-agent] > /tools
```

Tool	~Tokens	Permission
Built-in		
- read	870	trusted
aws-knowledge-mcp-server (MCP)		
- aws___read_documentation	900	not trusted
- aws___recommend	420	not trusted
- aws___search_documentation	2.2k	trusted
awslabs.aws-pricing-mcp-server (MCP)		
- analyze_cdk_project	110	not trusted
- analyze_terraform_project	110	not trusted
- generate_cost_report	1.4k	not trusted
- get_bedrock_patterns	60	not trusted
- get_price_list_urls	350	not trusted
- get_pricing	3.8k	trusted
- get_pricing_attribute_values	400	not trusted
- get_pricing_service_attributes	250	not trusted
- get_pricing_service_codes	200	not trusted

4. Test using aliases:

```
"Use aws_search to find Lambda documentation"
"Use pricing_lookup to get EC2 pricing"
"Search AWS documentation using aws_search for S3 best practices"
"Look up RDS pricing using pricing_lookup"
```

Verification

- ✔ Tool aliases work as expected
- ✔ Original tool names still function
- ✔ Aliases make complex tool names more user-friendly
- ✔ Both aliased and original names appear in tool lists

Key Learnings

- `toolAliases` creates user-friendly names for complex tool names
- Aliases are particularly useful for long MCP tool names
- Both original and aliased names remain functional
- Aliases improve user experience and workflow efficiency